Prabhavit Gupta



AI Architect / Solutions Engineer: Agentic Systems, RAG & API Orchestration

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Summary

AI solutions engineer and technical architect with 3 years of hands-on commercial experience designing, building, and deploying production-grade LLM applications, agentic workflows, and knowledge retrieval pipelines. As Lead Developer at FlowSpace, architected multi-step customer support agents with tool use and live API function calling, end-to-end RAG pipelines with vector databases for automated document parsing, Stripe payment processing integrations, and observability safeguards. Deep expertise directing LLM tool-calling architectures, structured JSON outputs, n8n automations, Claude Code workflows, and robust RESTful integrations with TypeScript and working-level Python. Based in Venice, Italy with an EU Long Term Residence Permit (unrestricted EU right to work without sponsorship). Immediately available.

Technical Expertise

- **Generative AI & Agents:** Multi-agent orchestration, Tool Calling, Function Calling, Prompt Engineering, Structured JSON Outputs, Agentic workflows, Model Context Protocol (MCP).
- **RAG & Retrieval:** Semantic search, Vector databases (pgvector, Supabase), Embeddings, Chunking, Hybrid retrieval, Reranking strategies.
- **Fintech & API Integrations:** Stripe payment gateways, Webhook listeners, Idempotent transaction processing, REST API architecture, OfficeRnD, Nexudus.
- **LLMOps & Guardrails:** LLM tracing, Observability, Token and cost optimization, Schema validation, Automated error handling, Human-in-the-loop review.
- **Software Engineering:** TypeScript, Node.js, Working-level Python, PostgreSQL, SQL, Docker, n8n, Playwright, Git, Postman.
- **Languages & Authorization:** Hindi (Native), English (C1 Professional), Italian (Beginner). EU Long Term Residence Permit (No EU sponsorship required).

Experience

FlowSpace.co **Mar 2024 - Aug 2026**
Lead Developer (Full-time, Freelance) *London, UK (remote)*

- Architected and shipped production customer support AI agents using LLM tool use and function calling against live platform APIs, evaluating confidence scores and escalating complex queries to staff.
- Designed an automated AI listing and knowledge ingestion pipeline with human-in-the-loop review: parsed structured property records from heterogeneous documents (PDFs, spreadsheets, unstructured pages) into PostgreSQL/Supabase, drafting 900+ verified listings.
- Architected financial and booking integrations: implemented secure Stripe payment flows, webhook transaction reconciliation, and bidirectional booking synchronization with OfficeRnD and Nexudus APIs.
- Directed architecture migration off a legacy stack to a modern Next.js, TypeScript, and Laravel backend, establishing API specifications and schema contracts across teams.
- Implemented token and cost monitoring, schema guardrails, and automated regression testing using Playwright to ensure high reliability across production endpoints.
- Scaled platform user base from 2,000 to 4,500+ active users through iterative architectural improvements and resilient backend automations.

Urlink **Apr 2025 - Jul 2025**
Project Manager & AI Assisted Developer (Freelance) *Barcelona, Spain (remote)*

- Led architecture and rapid technical delivery of AI-assisted web applications, coordinating API contracts, database schemas, and external service integrations.
- Implemented automated testing pipelines and CI/CD verification workflows, reducing deployment regressions and release cycle times.
- Evaluated external AI APIs and models for cost, latency, and output consistency, integrating optimal models for client-specific use cases.

Independent Technical Consultant **Aug 2023 - Feb 2024**
AI Solutions & Automation Engineer *Venice, Italy*

- Built customized AI automation workflows, web scrapers, and data enrichment pipelines for European SME clients using n8n, Playwright, and OpenAI APIs.
- Implemented idempotent CRM webhook handlers and automated invoice reconciliation workflows, eliminating manual data entry.
- Delivered custom RAG-powered chatbot prototypes integrated with company knowledge bases for automated inquiry triage.

Education

Ca' Foscari University of Venice **2020 - 2024**
BA in Philosophy, International Studies and Economics *Venice, Italy*

- Rigorous focus on formal logic, analytical problem solving, systems analysis, and macroeconomic decision models.